

UNIT I – COMPUTER VISION (Simple Notes)

1. Computer Vision Computer Vision is a part of Artificial Intelligence that helps computers see and understand images and videos like humans. It is used in face recognition, medical imaging, self-driving cars, and surveillance.

2. Photometric Image Formation and Digital Camera Photometric image formation explains how light falls on an object, reflects from it, and forms an image on a camera sensor. A digital camera uses a lens to focus light, a sensor (CCD/CMOS) to convert light into electrical signals, and stores the image digitally.

3. Geometric Primitives Geometric primitives are basic shapes used in images such as points, lines, curves, circles, and regions. They help in representing object shapes and structures.

4. Edge and Line Detection Edge detection finds boundaries in an image where brightness changes suddenly. Common methods are Sobel and Canny. Line detection finds straight lines using methods like Hough Transform.

5. Image Segmentation and Active Contours Image segmentation divides an image into meaningful parts. Active contour models (snakes) are curves that move towards object boundaries to get accurate shapes.

6. Feature Detectors and Feature Descriptors Feature detectors find important points in an image. Feature descriptors describe those points numerically. Both are used together for image matching and recognition.

7. Histogram Processing Histogram processing improves image contrast and brightness. Techniques like histogram equalization make images clearer.

8. Image Restoration and Global Optimization Image restoration improves degraded images by removing noise and blur. Global optimization methods find the best solution by minimizing errors.

9. Feature Matching and Feature Tracking Feature matching finds the same features in different images. Feature tracking follows features across video frames. They are used in video tracking, robotics, and augmented reality.